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8 September 1972
Superseding
MIL-R-38522A(USAF)
1 June 1971

# MILITARY SPECIFICATION
# RADOME CW-860/GPS( )

## 1. SCOPE

1.1 This specification covers one type of metal space frame radome, designated Radome CW-860/GPS( ), hereinafter referred to as the "radome".

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. In case of conflict between this specification and the following documents, this specification shall govern.

SPECIFICATIONS

Federal

| | |
|---|---|
| L-P-383 | Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated |
| QQ-A-200/8 | Aluminum Alloy Bar, Rod Shapes, Tube and Wire, Extruded, and Structural Shapes, 6061 |
| QQ-A-225/8 | Aluminum Alloy Bar, Rod, Wire and Special Shapes, Rolled, Drawn or Cold Finished 6061 |
| QQ-A-250 | Aluminum Alloy Plate and Sheet; General Specification For |
| QQ-A-250/11 | Aluminum Alloy 6061, Plate and Sheet |
| RR-L-91 | Ladder, Straight, Extension, and Trestle |

Military

| | |
|---|---|
| MIL-E-4158 | Electronic Equipment, Ground, General Requirements for |
| MIL-N-7513 | Nomenclature Assignment, Contractor's Method for Obtaining |
| MIL-R-7575 | Resin Polyester, Low-Pressure Laminating |
| MIL-C-9084 | Cloth, Glass, Finished, For Polyester Resin Laminates |
| MIL-R-9300 | Resin, Epoxy, Low-Pressure Laminating |
| MIL-P-9400 | Plastic Laminate Materials And Sandwich Construction, Glass Fiber Base, Low-Pressure Aircraft Structural, Process Specification Requirements |



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| | |
|---|---|
| MIL-R-17343 | Rope, Nylon |
| MIL-A-22152 | Aluminum Alloy Sand Castings Heat Treatment Processes For |
| MIL-P-25421 | Plastic Materials, Glass Fiber Base-Epoxy Resin, Low Pressure Laminated |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-406 | Plastic, Method of Testing |

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-831 | Test Reports, Preparation Of |

DRAWINGS

Air Force

| | |
|---|---|
| 53B54701 | Rope, Radome |
| 58C55560 | Bracket, Mounting |
| 58C55561 | Bumper |
| 58D55555 | Ladder Rigid Radome Outside Service |
| S-7078-2 | Base Bolt Circle Layout Radome CW-396(A) |

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM)

| | |
|---|---|
| A36 | Structural Steel, Specification For |
| A233 | Mild Steel Covered Arc-Welding Electrodes, Specification For |
| A235 | Carbon Steel Forgings For General Industrial Use, Specifications For |
| A237 | Alloy Steel Forgings For General Industrial Use, Specifications For |
| A307 | Low-Carbon Steel Externally and Internally Threaded Standard Fasteners, Specifications For |
| A325 | High Strength Bolts For Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers, Specifications For |



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| | |
|---|---|
| A354 | Quenched and Tempered Alloy Steel Bolts and Studs with Suitable Nuts, Specifications For |
| B184 | Aluminum and Aluminum-Alloy Arc-Welding Electrodes, Specifications For |
| D-635 | Flammability of Self-Supporting Plastics, Test For |
| D-757 | Flammability of Plastics, Self-Extinguishing Type, Test For |

(Copies of ASTM documents may be obtained form The American Society For Testing and Materials, 1916 Race St., Philadelphia, Pennsylvania, 19103).

American National Standards Institute (ANSI)

| | |
|---|---|
| G41.2 | High Strength Low-Alloy Structural Steel, Specifications For |
| G41.3 | High Strength Structural Steel, Specifications For |
| G41.4 | High Strength Low-Alloy Structural Manganese Vanadium Steel, Specifications For |
| G42.2 | Steel Structural Rivets, Specifications For |
| G50.1 | Mild-To-Medium-Strength Carbon-Steel Castings For General Application, Specifications For |
| G52.1 | High-Strength Steel Castings For Structural Purposes, Specifications For |

(Copies of ANSI documents may be obtained from the American National Standards Institute, 1430 Broadway, New York, New York 10018).

Department of Commerce

| | |
|---|---|
| FAA | Technical Standard Order TSO-N25 Obstruction Lighting Basic Requirement |
| FAA | Obstruction Marking And Lighting AGA-N53 |

(Copies of Federal Aviation Administration (FAA) documents may be obtained from Federal Aviation Administration, 800 Independence Ave, S. 20, Washington, D.C. 20553.)

Department of Labor

| | |
|---|---|
| OSHA | Part 1910 - Occupational Safety And Health Standards |



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(Copies of OSHA documents may be obtained from U.S. Government Printing Office, Washington, D.C.)

## 3. REQUIREMENTS

3.1 Preproduction.-This specification provides for preproduction testing.

3.2 Components.-The radome shall consist of:

| Item No. | Quantity | Description | Requirement |
|---|---|---|---|
| 1 | 1 each | Radome | 3.5.1 |
| 2 | 1 each | Lightning Protection assembly | 3.5.2 |
| 3 | As Required | Maintenance Equipment | 3.5.3 |
| 4 | 1 each | Repair kit | 3.5.4 |
| 5 | 1 each | Aircraft warning light system | 3.5.5 |

3.3 General Specification.-The requirements of MIL-E-4158 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3.1 Exceptions to requirements of MIL-E-4158.

3.3.1.1 Service Conditions.-The radome shall be designed and constructed to withstand any individual or probable combination of the service conditions specified below without any mechanical or electrical damage, or performance degradation:

| | |
|---|---|
| a. Low temperature | |
| (1) Storage, non-operation | -62°C (-80°F) |
| (2) Operation | -54°C (-65°F) |
| b. High temperature | |
| (1) Storage, non-operation | +71°C (+160°F) |
| (2) Operation | +71°C (+160°F) |
| c. Low pressure | |
| (1) Non-operation and storage | From sea level to 3.44 inches of mercury (50,000 ft) |
| (2) Operation | From sea level to 20.6 inches of mercury (10,000 ft) |
| d. Relative humidity | |
| (1) Non-operation and storage | Up to 100 percent including condensation due to temperature changes |



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| | |
|---|---|
| e. Sand and dust | As encountered in desert and other dust laden areas |
| f. Sunshine | As encountered at the earth's surface |
| g. Fungus | As encountered in warm humid atmospheres |
| h. Salt fog | As encountered in the salt atmosphere of ocean transportation and in coastal areas |
| i. Rain | As encountered in monsoon areas |
| j. Wind load | 250 knots (230 miles per hour) |
| k. Snow and ice | 3 inches of rime ice or snow or combination thereof measured over the top quarter of the radome. |
| l. Shock | As encountered during handling, transportation, and service use. |
| m. Vibration | As encountered during shipment by land, sea or air. |

3.3.1.2 Material.-All materials used in the fabrication of the radome shall appear on appropriate Government "Qualified Products List" or shall be certified as being in accordance with appropriate military specification by an independent certified test laboratory. All materials shall be subject to the approval of the procuring activity when so directed by the contract. (See 6.2.f)

3.3.1.3 Base fabric.-The base fabric reinforcement shall be fiberglass in accordance with the requirements of MIL-P-9084 or dacron, polypropylene or other materials subject to the approval of the contracting agency.

3.3.1.4 Resin.-The resin selected shall be in accordance with the requirements of MIL-R-7575 Type III or MIL-R-9300 Class III.

3.3.1.5 Laminates.-Laminated skins shall be in accordance with the requirements of Federal Specification L-P-383, or MIL-P-25421, Class III. Inspectional requirements as outlined in MIL-P-9400 shall apply to laminated skins. The laminate shall be fire resistant in accordance with ASTM D-635 and ASTM D-757. The burning rate specified in ASTM D-757 shall not exceed 0.2 inch per minute. Mechanical testing of laminates shall be conducted in accordance with FED-STD-406.



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3.3.1.6 Adhesives.-All adhesives shall be of the best commercial grade consistent with the physical, electrical and operating requirements specified herein.

3.3.1.7 Structural aluminum.-Structural aluminum utilized in the construction of the radome shall be in accordance with QQ-A-200/8, QQ-A-225/8, QQ-A-250, QQ-A-250/11 or MIL-A-22152.

3.3.1.8 Structural steel.-Structural steel utilized in the construction of the radome shall be in accordance with ANSI and ASTM Standard G50.1 (Grade 65-35), A36, G52.1 (Grade 80-50), A235 (Class C1, F, and G), A237 (Class A), G41.2, G41.3 or G41.4.

3.3.1.9 Hardware.-All necessary hardware such as bolts, nuts, washers or rivets required for panel assembly, radome assembly or installation of the radome to the existing mounting ring shall be furnished in accordance with ASTM and ANSI Standard G42.2, A325, A354 (Grade BC) or A307.

3.3.1.10 Welding.-Steel welding shall be in accordance with ASTM Standard A36. Electrodes for steel and aluminum welding shall be in accordance with ASTM Standards A233 and B184 respectively. All welding techniques and quality of welds made shall conform to the standards of the American Welding Society.

3.4 Design

3.4.1 Design Analysis.-When so required by the contract, prior to the ordering of material, the contractor shall submit to the procuring activity two complete copies of a design analysis which demonstrates by charts, graphs, tables, drawings or calculation that the radome will satisfactorily meet the requirements of 3.4. The calculations shall be presented in a clear and legible form and shall include at least the following: (See 6.2.c)

a. Overall inside clear radome dimensions

b. Radome geometry

c. Panel breakdown by member and type

d. Number of members and average length

e. Average panel weight and area

f. Membrane attachment details

g. Cross-sectional dimensions of members and membranes

h. Materials and their required mechanical and electromagnetic characteristics

i. Hardware details

j. Base attachment details



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All loading conditions, formulae and references shall be identified as to the source. The design analysis shall be insofar as possible a complete document not requiring extensive use of reference material. Any reference material of a proprietary nature or not available through normal channels shall accompany the design analysis. In addition, a separate electrical analysis shall be performed to demonstrate that the requirements of paragraph 3.4.2 are met.

3.4.2 Panel coating.-The exterior surface of the radome panels shall be provided with an integrally bonded film of Tedlar (Polyvinylflouride). The Tedlar film shall be of sufficient thickness to provide a maintenance free life of 10 years. The interior surface of the radome panels shall not be coated.

3.4.3 Process specification.-The contractor shall provide a complete process specification for the fabrication of the panels, sections or any other component detailing all steps in the manufacture of the items identifying all materials including metals (either by description or trade or chemical name and source or both) and describing all equipment necessary to the fabrication. The materials and processes in production shall conform to the requirements of the process specifications, in addition to the requirements of this specification. After approval of the process specification, no changes in the production procedure shall be made without approval of a new superseding process specification. The process specification shall include the requirements of MIL-P-9400. The process specification shall be submitted to the procuring activity when so required by the contract. (See 6.2.d)

3.5 Details of Components

3.5.1 Radome.-The radome shall be a metal space frame structure in the shape of a faceted or smooth sphere approximately 55 feet in major diameter and truncated to a height of approximately 39.2 feet. The radome shall be designed and manufactured to be installed on the base bolt circle shown on Drawing S-7078-2 without the aid of an adapter ring and without a modification of the existing base ring. The entire structure shall be capable of withstanding the specified service conditions without the aid of additional trusses, guys or supports.

3.5.1.1 Radome panels.-The term "radome panel" shall refer to either a membrane bounded by the metal frame members as a unit or a membrane alone which is subsequently attached to the completed radome framework. The radome panels shall be designed and constructed for assembly with simple hardware. The panels shall not be permanently fastened to the radome by application of adhesive cements or other bonding materials over the total flange surface which would prevent removal of any individual panel or group of panels at any time. The panels size shall be consistent with good engineering design judgement exercised in considering maintenance and erection in adverse environmental conditions.

3.5.1.1.1 Panel joints.-Panel joints shall present a smooth exterior to the complete radome and shall be suitably caulked to prevent free entry of moisture, water or both. Weatherproof caulking compound shall be provided with each radome in sufficient quantity so that when applied to the erected radome there will be no passage of water from the outside to the inside of the radome.



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3.5.1.1.2 Panel assembly hardware.-Panel assembly hardware shall be supplied with each radome in accordance with 3.3.1.9. Hardware materials and finishes not specified shall be in accordance with applicable ASTM, ASA Standards or military specifications.

3.5.1.1.3 Panel coating.-The exterior surface of the radome panels shall be provided a protective finish in accordance with 3.4.2.

3.5.1.2 Base attachment hardware.-Each radome shall be provided with sufficient 1/2 inch bolts, nuts and washers in accordance with 3.3.1.9 to accommodate the base bolt pattern shown on Drawing S-7078-2.

3.5.2 Lightning protection assembly. A suitable lightning protection assembly shall be provided for installation at the apex of the radome. The system shall be in the proper size and material to provide protection to the warning light system. Adequate means shall be provided to ground the systems to the metal frame of the radome.

3.5.3 Maintenance and erection equipment. Each radome shall be furnished with the following maintenance and erection equipment which shall be subject to the approval of the procuring activity: (See 6.2.e)

a. Outside service ladder. The outside service ladder shall be made of magnesium or aluminum and shall be of the sectional hinged type in accordance with Drawing 58D55555. It shall consist of nine sections.

b. Interior service ladders. The interior service ladders shall consist of one 12-foot aluminum ladder (RR-L-0091, Type 1, Grade B, Kind 1, with Safety Shoes S1) and one 52-foot aluminum extension ladder (RR-L-0091, Type IIb, Grade B, Kind 1, with Safety Shoes S1). In addition to safety shoes to prevent ladder slipping, both ladders shall be fitted with bearing plates to protect the radome. These bearing plates shall be composed of one each Bumper, Drawing 58C55561, and one each Bracket Mounting, Drawing 58C55560, adapted to fasten to side rails near top of ladder.

c. Pulley block and tackle. The pulley block and tackle shall consist of four single sheave blocks and 200 feet of 1/2-inch nylon rope per MIL-R-17343.

d. Snow rope. A nylon rope to be attached to the lightning protection assembly shall be provided. The rope shall be in accordance with Drawing 53B54701 and shall be 125 feet in length.

e. Safety belt. Two each seat type trimmers safety belt, Miller Equipment Co, model 6625, shall be provided.

f. Ring and pulley assembly. An interior ring and exterior pulley (horizontally mounted) shall be permanently mounted to the apex of the radome as part of the Lightning Protection Assembly. Each shall be capable of individually supporting 2000 pounds (not simultaneously).

3.5.3.1 Radome erection equipment. When required by the contract, the following radome erection equipment shall be provided (See 6.2.n). The radome erection equipment shall consist of demountable metal scaffolds having the following characteristics. All scaffolding shall meet the requirements of OSHA standard.



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a. Two ladder type scaffolds 31 to 32 feet high (adjustable) which may be joined together to make a stable working platform or used separately. Each scaffold shall have outrigger support legs for stability.

b. Base sections of each scaffold and all outrigger support legs shall be equipped with 8-inch diameter, rubber-tired, lockable swivel casters.

c. Two outboard support platforms shall be provided with each scaffold to enable workers to safely reach and work on any interior area of the spherical radome. Radome is a truncated sphere of 55-foot diameter with a base diameter at the truncated section of 50 feet.

d. A top working platform shall be provided which will connect the two scaffolds and provide a stable working platform of at least 4' X 26' at the 31 to 32 foot level. When joined together in this manner, the two scaffolds shall present an essentially clear opening of 12' X 26' minimum which may be crossed by stiffeners or spreaders. Such spreaders shall be easily relocatable and widely adjustable to clear the particular antenna structure being straddled.

e. All work platforms shall be equipped with guard rails at a height of 36 to 42 inches except where such rails may have to be reduced in height near the ends to avoid interference with radome geometry.

f. Each of the scaffolds shall be furnished with an easily relocatable "T" bar hoist to facilitate lifting of radome panels, scaffold sections, tools and material from the deck to the working level. Hoists shall be equipped with two pulleys capable of accepting 1/4-inch diameter fiber rope for lifting.

g. Complete assembly, disassembly and packing instructions shall be furnished with this equipment. Equipment furnished under this paragraph shall be packed separately from radome in reusable containers to facilitate its storage and/or reshipment.

3.5.4 Repair kit. When required by contract (See 6.2), a suitable repair kit consisting of the necessary materials, adhesives and tools for the minor repair of the radome membranes in the field shall be provided as part of the radome. Resins shall be of the activated or promoted type to reduce the hazard of packaging extremely reactive activators and catalysts in the same kit even though this materially reduces the shelf life of the resins. Resin must be compatible with that used in manufacture of the radome and meet same specifications with respect to physical and electrical characteristics. Material that has a short shelf life shall be suitably dated, so that it can be replaced at certain intervals. All materials which have a shelf life and/or pose any sort of handling threat to personnel engaged in storage or use shall be appropriately labelled with warnings, instructions and action to be taken with regard to personnel injured by them. Such information shall be on individual containers as well as the larger packages or kits containing them. The kit shall be packaged in a suitable wooden box and shipped together with the radome. Each repair kit shall contain the necessary instructions for making minor repairs.



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3.5.5 Aircraft warning lights. When required by the contract, the contractor shall provide a complete aircraft warning light assembly in accordance with TSO-N25 and AGA-N53 of the Department of Commerce, with the exception that two lights shall be installed. One light shall be operational at any time. The second light shall be automatically relayed into operation in the event of failure of the first light. The lighting system shall be service free for a minimum of one year. The assembly shall include the wiring and conduit to the base of the radome (See 6.2.m).

3.6 Performance. The radome, as designed, manufactured and installed shall meet the following performance requirements.

3.6.1 Electromagnetic performance. The radome shall not degrade the radiation characteristics of the enclosed antenna system beyond the following limits over a passband from 1.0 to 6.0 Gigahertz (GHz) at either horizontal or vertical polarization.

| Radiation Characteristics | Allowable Degradation |
|---|---|
| Side lobe level without radome | Average side lobe level (see Note 1) increase with radome |
| -20 decibels (DB) | +0.5 DB |
| -25 DB | +1.0 DB |
| -30 DB | +2.0 DB |
| -35 DB | +3.0 DB |
| Antenna gain | Loss with radome shall not exceed 0.5 DB one way |
| Beamwidth | Changes in beamwidth with radome shall not exceed 5.0 percent |
| Boresight | Boresight error with radome shall not exceed 0.1 milliradian |
| Boresight error rate | Boresight error rate with radome shall not exceed 0.01 mils per mil per degree |
| Voltage Standing Wave Ratio (VSWR) | Change in VSWR with radome shall not exceed 5 percent |

Note 1: The average increase in side lobe level shall be computed for each antenna pattern recorded. The averaging procedure shall consist of calculating peak-to-peak side lobe level increases (due to the radome) within each 2 degree interval of a pattern and computing the average of these changes over the entire 360 degrees of the pattern.



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3.6.2 Maintainability. The individual radome panel membranes shall be capable of being repaired in place with the kit specified in 3.5.4 when penetrated by wind borne debris. The time to replace a single panel anywhere in the radome, with the equipment specified in 3.5.4, shall not exceed one hour.

3.6.3 Service life. The equipment furnished under this specification shall have a service life of 10 years with normal maintenance.

3.7 Nomenclature. When so required by the contract, nomenclature shall be obtained in accordance with the requirements of MIL-N-7513. The parenthesis appearing in the nomenclature may be dropped, or replaced by a letter, to designate the particular design. (See 6.2.j)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.g)

4.2 Classification of tests. The inspection and testing of the radome shall be calssified as follows:

a. Preproduction tests . . . . . (See 4.4)

b. Acceptance tests . . . . . (See 4.5)

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 Preproduction tests

4.4.1 Preproduction test samples. The preproduction test samples shall consist of reinforced plastics, laminates and metallic members prepared in accordance with drawings and with the process specification. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. (See 6.2.i)

4.4.2 Preproduction test reports. When so required by the contract, the contractor shall, after completion of the tests, prepare a preproduction test report in accordance with the requirements of MIL-STD-831, and shall furnish three copies of the report to the procuring activity. (See 6.2.h)

4.4.3 Preproduction tests. The preproduction tests shall consist of the following tests:



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a. Material certification. The contractor shall conduct the tests necessary to demonstrate that the radome materials possess the mechanical and electromagnetic characteristics required by the design analysis and appropriate military specifications. Laminate tests shall be conducted in accordance with the requirements of FED-STD-406. Where the contractor is purchasing finished materials, certification from the supplier will be acceptable in lieu of testing.

b. Quality control panel. At the outset of production the procuring activity shall select a representative panel, fabricated in accordance with the process specification, to be preserved as a visual quality standard for the remainder of the contract. This panel shall be representative of the minimum acceptable quality of workmanship and shall serve as the standard for visual comparison with subsequent panels. The standard for dimensional inspection shall at all times be the drawings.

c. Pressurization test. Prior to production, the contractor shall fabricate a panel (largest in area used in radome) in accordance with the process specification. The panel shall be suitably secured and the membrane subjected to a negative pressure of 0.73 pounds per square inch (PSI). This pressure shall be held for 30 seconds and released. The cycle shall be repeated 10 times and the panel examined for any indication of failure.

4.5 Acceptance tests. The radome shall be subjected to acceptance tests. The acceptance tests shall consist of the following tests as described under 4.6 "Test methods" of this specification.

4.6 Test methods

4.6.1 Mechanical inspection tests. The radome shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Fit of base course to the bolt circle layout

b. Fit and ease of assembly of panels with particular attention to ease of bolting

c. Soundness of welds after assembly

d. Tightness of joints

e. Nameplates, identification markings and labels

f. Other visual defects

g. Fit of lightning protection assembly, aircraft warning light, and outside service ladder

h. Operation of outside service ladder



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4.6.2 Material certification. For each lot of material ordered or radome manufactured, the contractor, when so required by the contract, shall submit supplier certification or test data to the procuring activity demonstrating the material characteristics are consistent with the requirements of the design analysis. (See 6.2.k)

4.6.3 Pressurization. Two panels shall be tested in accordance with the procedure outlined in paragraph 4.4.3.c.

4.6.4 Visual inspection. The radome components shall be given a thorough visual inspection to determine if the workmanship is in accordance with the quality control panel, the process specification and the requirements of this specification.

4.6.5 Dimensional inspection. A program shall be established for screening a suitable percentage of panels or components from each radome for dimensional check. The standard for dimensional inspection shall at all times be the drawings.

4.6.6 Environmental tests. The radome shall meet all specified requirements when subjected to the following sequence of environmental tests in accordance with MIL-STD-810:

a. Low temperature: Method 502, Procedure I. Step 2 storage temperature -62°C (-80°F) for 48 hours; Step 4 operation temperature -54°C (-65°F).

b. High temperature: Method 501, Procedure II. Step 7 operating temperature +71°C (+160°F).

c. Low pressure: Method 500, Procedure I, Step 3 10,000 ft (20.6 inches of mercury).

d. Sunshine: Method 505, Procedure I. Test purpose to determine degradation and deterioration.

e. Dust: Method 510, ProcedureI. Test Steps 2 and 3 to run consecutively.

f. Shock: Method 516, Procedure I (40g peak value with 18 milliseconds duration) and Procedure V.

g. Vibration: Method 514, Procedure X (Curve AB of Figure 514-6), and Procedure XI (Part 2).

h. Rain: Method 506, Procedure I.

i. Humidity: Method 507, Procedure I.

j. Fungus: Method 508, Procedure I.

k. Salt fog: Method 509, Procedure I.

l. Wind, snow and ice test: The requirements specified in paragraphs 3.3.1.1(j) and (k) will be proven by submission of a detail structural analysis. This will be subject to the approval of the procuring activity. (See 6.2.l)



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4.6.7 Erection test. One radome shall be completely erected using the erection equipment provided with the radome. Dimensional measurements shall be made to determine if they are within the specification requirements. The fit of the individual panels within the radome shall be observed to determine that they fit properly and there are no gaps at the joints which might cause water leakage. (This observation is to be made after the bolts are torqued to the specified value).

## 5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity. The equipment shall be packed for overseas shipment in accordance with the instructions of the procuring activity.

5.2 Marking. The equipment shall be marked for shipping and storage in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use. Radome CW-860/GPS( ) is intended for use on Towers, AB-259/FPS-6, AB-259A/FPS-6 and AB-536/GPS as a protective cover for either the antenna of radar sets such as AN/FPS-3, AN/FPS-20, AN/FPS-7, AN/FPS-6, AN/FPS-26 or AN/FSS-7.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable level of preservation, packaging and packing required. (See 5.1)

c. Conditions under which design analysis approval will be granted. (See 3.4.1)

d. When process specification shall be submitted. (See 3.4.3)

e. Conditions under which approval will be granted for the required maintenance and erection equipment furnished by the contractor. (See 3.5.3)

f. When materials shall be submitted for approval. (See 3.3.1.2)

g. Whether testing will be conducted by the procuring activity or by the contractor. (See 4.1)

h. Whether a preproduction test report will be submitted by the contractor. (See 4.4.2)

i. When preproduction test samples shall be submitted. (See 4.4.1)

j. Whether nomenclature and nameplate data will be required of the contractor and when this data will be provided to the procuring activity. (See 3.7)

k. Whether the contractor will submit a material certification. (See 4.6.2)



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l. Conditions under which approval will be granted to the contractor by the procuring activity for these tests. (See 4.6.6(l))

m. Whether aircraft warning lights are required. (See 3.5.5)

n. Whether radome erection equipment is to be provided with this radome.

o. Whether repair kit is required. (See 3.5.4)

6.3 Any additional information required for design characteristics will be furnished, if available, to the contractor upon request to the procuring activity.

Custodian:
Air Force - 80

Preparing Activity:
Air Force - 80
Project 5985-(F)369



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *Printed or typed name and activity - Optional*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

S/N 0102-014-1801 0-25254

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